

Problem 1

Given the vector functions:

$$\mathbf{r}_u = \begin{pmatrix} R \cos u \cos v \\ R \cos u \sin v \\ -R \sin u \end{pmatrix}, \quad \mathbf{r}_v = \begin{pmatrix} -R \sin u \sin v \\ R \sin u \cos v \\ 0 \end{pmatrix}$$

$$\mathbf{r}_{uu} = \begin{pmatrix} -R \sin u \cos v \\ -R \sin u \sin v \\ -R \cos u \end{pmatrix}, \quad \mathbf{r}_{uv} = \begin{pmatrix} -R \cos u \sin v \\ R \cos u \cos v \\ 0 \end{pmatrix}, \quad \mathbf{r}_{vv} = \begin{pmatrix} -R \sin u \cos v \\ -R \sin u \sin v \\ 0 \end{pmatrix}$$

Computing the cross product:

$$\begin{aligned} \mathbf{r}_u \times \mathbf{r}_v &= R^2 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos u \cos v & \cos u \sin v & \sin u \\ -\sin u \sin v & \sin u \cos v & 0 \end{vmatrix} \\ &= R^2 \begin{pmatrix} \sin^2 u \cos v \\ \sin^2 u \sin v \\ \sin u \cos u \end{pmatrix} \end{aligned}$$

The magnitude is

$$|\mathbf{r}_u \times \mathbf{r}_v| = R^2 \sqrt{\sin^4 u + \sin^2 u (1 - \sin^2 u)} = R^2 \sin u$$

Thus, the normal vector $\mathbf{n}$ is

$$\mathbf{n} = \begin{pmatrix} \sin u \cos v \\ \sin u \sin v \\ \cos u \end{pmatrix}$$

Calculating the First Fundamental Form coefficients:

$$\begin{aligned} E &= R^2(\cos^2 u + \sin^2 u) = R^2 \\ F &= 0 \\ G &= R^2 \sin^2 u \end{aligned}$$

Therefore, the determinant is

$$|I| = R^4 \sin^2 u$$

Calculating the Second Fundamental Form coefficients

$$\begin{aligned} e &= -R \sin^2 u \cos^2 v - R \sin^2 u \sin^2 v - R \cos^2 u = -R \\ f &= -R \sin u \cos u \sin v \cos v + R \sin u \cos u \sin v \cos v = 0 \\ g &= -R \sin^2 u \cos^2 v - R \sin^2 u \sin^2 v = -R \sin^2 u \end{aligned}$$

Therefore, the determinant is

$$|II| = R^2 \sin^2 u$$

Finally, the Gaussian Curvature K is

$$K = \frac{|II|}{|I|} = \frac{1}{R^2}$$

Problem 2(a)

We start with the Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\varphi}{dx^2} = E\varphi$$

Using the substitution $p' + p \frac{\varphi'}{\varphi} = \frac{\varphi''}{\varphi} \implies p' + p^2 = \frac{\varphi''}{\varphi}$, we substitute into the main equation:

$$\begin{aligned} -\frac{\hbar^2}{2m}(p' + p^2) &= E \\ p' + p^2 &= -\frac{2mE}{\hbar^2} \end{aligned}$$

Let $k^2 = \frac{2mE}{\hbar^2}$. The equation becomes:

$$\begin{aligned} \frac{dp}{p^2 + k^2} &= -dx \\ \frac{1}{k} \tan^{-1}\left(\frac{p}{k}\right) &= -x + C \\ p &= -k \tan(kx + C_1) \end{aligned}$$

Since $p = \frac{\varphi'}{\varphi}$,

$$\begin{aligned} \frac{\varphi'}{\varphi} &= -k \tan(kx + C_1) \\ \frac{d\varphi}{\varphi} &= -k \tan(kx + C_1) dx \\ \ln \varphi &= \ln[\cos(kx + C_1)] + C_2 \\ \varphi &= C_3 \cos(kx + C_1) \end{aligned}$$

Problem 2(b)

The characteristic equation is

$$-\lambda^2 = k^2 \implies \lambda = \pm ki$$

By definition, the general solution is

$$\varphi = A_1 \sin(kx) + B_1 \cos(kx)$$

Alternatively written as

$$\varphi = A_2 \cos(kx + \theta)$$

which matches the form derived in part (a).

Problem 2(c)

Applying boundary conditions:

1. $\varphi(0) = 0 \implies B_1 = 0.$
2. $\varphi(L) = 0 \implies \sin(kL) = 0.$

This implies

$$kL = n\pi \implies \frac{2mE}{\hbar^2} = \frac{n^2\pi^2}{L^2}$$

Solving for E :

$$E = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

Problem 3

Given:

$$\hat{S}_n |\varphi_+\rangle = \frac{\hbar}{2} |\varphi_+\rangle$$

This expands to the matrix equation:

$$\sin \theta \cos \phi \begin{pmatrix} \beta \\ \alpha \end{pmatrix} + \sin \theta \sin \phi \begin{pmatrix} -i\beta \\ i\alpha \end{pmatrix} + \cos \theta \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Rearranging into a system of equations:

$$\begin{pmatrix} \cos \theta - 1 & \sin \theta \cos \phi - i \sin \theta \sin \phi \\ \sin \theta \cos \phi + i \sin \theta \sin \phi & -\cos \theta - 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We can express the components as

$$\begin{aligned} \alpha &= i \sin \theta \sin \phi - \sin \theta \cos \phi \\ \beta &= \cos \theta - 1 \end{aligned}$$

The norm is

$$\text{Norm} = \sqrt{\cos^2 \theta - 2 \cos \theta + 1 - \sin^2 \theta \sin^2 \phi - 2i \sin^2 \theta \sin \phi \cos \phi + \sin^2 \theta \cos^2 \phi}$$

Let the norm be denoted by k . The normalized state is

$$|\varphi_+\rangle = \frac{i \sin \theta \sin \phi - \sin \theta \cos \phi}{k} |+\rangle + \frac{\cos \theta - 1}{k} |-\rangle$$

Problem 4

The cross ratio is given by

$$[T(z_1), T(z_2); T(z_3), T(z_4)] = \frac{[T(z_1) - T(z_3)][T(z_2) - T(z_4)]}{[T(z_1) - T(z_4)][T(z_2) - T(z_3)]}$$

First, calculate the difference term:

$$\begin{aligned} T(z_1) - T(z_3) &= \frac{az_1 + b}{cz_1 + d} - \frac{az_3 + b}{cz_3 + d} \\ &= \frac{(az_1 + b)(cz_3 + d) - (az_3 + b)(cz_1 + d)}{(cz_1 + d)(cz_3 + d)} \\ &= \frac{(acz_1z_3 + adz_1 + bcz_3 + bd) - (acz_3z_1 + adz_3 + bcz_1 + bd)}{(cz_1 + d)(cz_3 + d)} \\ &= \frac{ad(z_1 - z_3) - bc(z_1 - z_3)}{(cz_1 + d)(cz_3 + d)} \\ &= \frac{(ad - bc)(z_1 - z_3)}{(cz_1 + d)(cz_3 + d)} \end{aligned}$$

Similarly, for any i, j , we have $T(z_i) - T(z_j) = \frac{(ad-bc)(z_i-z_j)}{(cz_i+d)(cz_j+d)}$.

Substituting this into the cross ratio expression:

$$\begin{aligned}
 \text{Cross Ratio} &= \frac{\left[\frac{(ad-bc)(z_1-z_3)}{(cz_1+d)(cz_3+d)} \right] \left[\frac{(ad-bc)(z_2-z_4)}{(cz_2+d)(cz_4+d)} \right]}{\left[\frac{(ad-bc)(z_1-z_4)}{(cz_1+d)(cz_4+d)} \right] \left[\frac{(ad-bc)(z_2-z_3)}{(cz_2+d)(cz_3+d)} \right]} \\
 &= \frac{(ad-bc)^2(z_1-z_3)(z_2-z_4)}{(cz_1+d)(cz_3+d)(cz_2+d)(cz_4+d)} \cdot \frac{(cz_1+d)(cz_4+d)(cz_2+d)(cz_3+d)}{(ad-bc)^2(z_1-z_4)(z_2-z_3)} \\
 &= \frac{(z_1-z_3)(z_2-z_4)}{(z_1-z_4)(z_2-z_3)} \\
 &= [z_1, z_2; z_3, z_4]
 \end{aligned}$$

Note: The $(ad-bc)^2$ terms cancel out, as do all the (cz_k+d) factors in the denominators, proving the cross ratio is invariant under Möbius transformations.

Problem 5(a)

Using the identity:

$$\tan x = -i \frac{e^{ix} - e^{-ix}}{e^{ix} + e^{-ix}} = -i \frac{e^{2ix} - 1}{e^{2ix} + 1}$$

We calculate:

$$\tan \frac{\pi}{12} = -i \frac{e^{i\pi/6} - 1}{e^{i\pi/6} + 1} = -i \frac{\sqrt{3}/2 + i/2 - 1}{\sqrt{3}/2 + i/2 + 1} = 2 - \sqrt{3}$$

Problem 5(b)

Note that:

$$2 - \sqrt{3} = \frac{(2 - \sqrt{3})(2 + \sqrt{3})}{2 + \sqrt{3}} = \frac{4 - 3}{2 + \sqrt{3}} = \frac{1}{2 + \sqrt{3}}$$

We rewrite the given equation by dividing both sides by $(2 + \sqrt{3})$:

$$\begin{aligned}
 \sin x + \frac{1}{2 + \sqrt{3}} \cos x &= \frac{\sqrt{2 + \sqrt{3}}}{2 + \sqrt{3}} \\
 \sin x + (2 - \sqrt{3}) \cos x &= \frac{1}{\sqrt{2 + \sqrt{3}}}
 \end{aligned}$$

Substitute $\tan \frac{\pi}{12}$ for $(2 - \sqrt{3})$:

$$\begin{aligned}\sin x + \tan\left(\frac{\pi}{12}\right) \cos x &= \sqrt{2 - \sqrt{3}} \\ \sin x + \frac{\sin(\pi/12)}{\cos(\pi/12)} \cos x &= \sqrt{2 - \sqrt{3}}\end{aligned}$$

Multiply through by $\cos \frac{\pi}{12}$:

$$\sin x \cos \frac{\pi}{12} + \cos x \sin \frac{\pi}{12} = \sqrt{2 - \sqrt{3}} \cos \frac{\pi}{12}$$

Using the sine addition formula on the LHS and simplifying the RHS using exact values:

$$\sin\left(x + \frac{\pi}{12}\right) = \left(\frac{\sqrt{3}-1}{\sqrt{2}}\right) \left(\frac{\sqrt{6}+\sqrt{2}}{4}\right) = \frac{1}{2}$$

Solving $\sin\left(x + \frac{\pi}{12}\right) = \frac{1}{2}$:

$$\begin{aligned}x + \frac{\pi}{12} &= n\pi + (-1)^n \frac{\pi}{6} \\ x &= n\pi + (-1)^n \frac{\pi}{6} - \frac{\pi}{12}, \quad n \in \mathbb{Z}\end{aligned}$$

Problem 6(a)

The velocity of ball 2 relative to ball 1 is:

$$\mathbf{v} = \begin{pmatrix} 5 \cos \theta - 1 \\ 5 \sin \theta \end{pmatrix}$$

Based on the geometry of intersection:

$$\begin{aligned} \frac{5 \sin \theta}{5 \cos \theta - 1} &= \frac{1}{2} \\ 10 \sin \theta &= 5 \cos \theta - 1 \\ 100(1 - \cos^2 \theta) &= 25 \cos^2 \theta - 10 \cos \theta + 1 \\ 125 \cos^2 \theta - 10 \cos \theta - 99 &= 0 \end{aligned}$$

Solving the quadratic:

$$\cos \theta = \frac{10 + 40\sqrt{31}}{250} = \frac{1 + 4\sqrt{31}}{25}$$

Thus, $\theta = \cos^{-1} \left(\frac{1+4\sqrt{31}}{25} \right)$. (The negative root is rejected as the ball is in the first quadrant).

Alternative Method: Using position vectors $\mathbf{r}_1 = \begin{pmatrix} 20+t \\ 10 \end{pmatrix}$ and $\mathbf{r}_2 = \begin{pmatrix} 5t \cos \theta \\ 5t \sin \theta \end{pmatrix}$:

$$\begin{cases} 20 + t = 5t \cos \theta \\ 10 = 5t \sin \theta \end{cases}$$

Squaring and adding equations to eliminate θ :

$$\begin{aligned} (20 + t)^2 + 100 &= 25t^2 \\ 24t^2 - 40t - 500 &= 0 \\ t &= \frac{5(1 + \sqrt{31})}{6} \quad (\text{reject } t < 0) \end{aligned}$$

Substituting back yields the same angle θ .

Problem 7(a)

Given $f'(x) = -\frac{3}{4}(625 + x^4)^{-7/4}(4x^3) = -\frac{3x^3 f(x)}{625 + x^4}$. Rearranging yields:

$$(625 + x^4)f'(x) + 3x^3 f(x) = 0$$

Problem 7(b)

We apply the Leibniz rule $(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(k)} v^{(n-k)}$.

Term 1: $((625 + x^4)f'(x))^{(n)}$ with $u = 625 + x^4, v = f'(x)$:

- $k = 0: (625 + x^4)f^{(n+1)}(x)$
- $k = 1: 4nx^3 f^{(n)}(x)$
- $k = 2: 6n(n-1)x^2 f^{(n-1)}(x)$
- $k = 3: 4n(n-1)(n-2)x f^{(n-2)}(x)$
- $k = 4: n(n-1)(n-2)(n-3)f^{(n-3)}(x)$

Term 2: $(3x^3 f(x))^{(n)}$ with $u = 3x^3, v = f(x)$:

- $k = 0: 3x^3 f^{(n)}(x)$
- $k = 1: 9nx^2 f^{(n-1)}(x)$
- $k = 2: 9n(n-1)x f^{(n-2)}(x)$
- $k = 3: 3n(n-1)(n-2)f^{(n-3)}(x)$

Grouping like derivatives and defining coefficients:

$$\begin{aligned} A_n &= 4n + 3 \\ B_n &= 3n(2n + 1) \\ C_n &= n(n-1)(4n + 1) \\ D_n &= n^2(n-1)(n-2) \end{aligned}$$

The equation becomes:

$$(625 + x^4)f^{(n+1)}(x) + A_n x^3 f^{(n)}(x) + B_n x^2 f^{(n-1)}(x) + C_n x f^{(n-2)}(x) + D_n f^{(n-3)}(x) = 0$$

Problem 7(c)

Substitute $x = 0$. All terms with x vanish:

$$\begin{aligned} 625f^{(n+1)}(0) + D_n f^{(n-3)}(0) &= 0 \\ f^{(n+1)}(0) &= -\frac{n^2(n-1)(n-2)}{625} f^{(n-3)}(0) \end{aligned}$$

Problem 7(d)

Base cases: $f^{(0)}(0) = \frac{1}{125}$, and $f^{(1)}(0) = f^{(2)}(0) = f^{(3)}(0) = 0$.

Calculating higher derivatives:

- For $n = 3$: $f^{(4)}(0) = -\frac{18}{78125}$
- For $n = 7$: $f^{(8)}(0) = \frac{5292}{9765625}$
- For $n = 4$: $f^{(9)}(0) \propto f^{(1)}(0) = 0$

Problem 7(e)

Vanishing Coefficients: Since $f^{(1)}(0) = f^{(2)}(0) = f^{(3)}(0) = 0$, $c_k = 0$ for $k \neq 4m$.

Recurrence: Let $n + 1 = 4m$.

$$c_{4m} = -\frac{4m-1}{2500m} c_{4m-4}$$

General Formula:

$$c_{4m} = \frac{1}{125} \left(-\frac{1}{2500} \right)^m \prod_{j=1}^m \frac{4j-1}{j}$$

Problem 8

For the loop:

$$\mathbf{B}_1 = \frac{\mu_0 I}{4\pi} \int_L \frac{d\mathbf{l}}{r^2} \hat{\mathbf{z}} = \frac{\mu_0 I}{4\pi} \int_0^\pi \frac{R d\theta}{R^2} \hat{\mathbf{z}} = \frac{\mu_0 I}{4R} \hat{\mathbf{z}}$$

For the two straight wires (using symmetry and limit integration):

$$\mathbf{B}_2 = \frac{\mu_0 I}{2\pi} \lim_{k \rightarrow \infty} \int_0^k \frac{R dy}{(R^2 + y^2)^{3/2}} \hat{\mathbf{z}} = \frac{\mu_0 I}{2\pi R} \hat{\mathbf{z}}$$

Total field:

$$\mathbf{B} = \frac{\mu_0 I}{R} \left(\frac{1}{4} + \frac{1}{2\pi} \right) \hat{\mathbf{z}}$$

Problem 9

Define the function and its derivative:

$$f(r) = \frac{M_S}{(R-r)^2} - \frac{M_E}{r^2} - \frac{M_S(R-r)}{R^3}$$

$$f'(r) = \frac{2M_S}{(R-r)^3} + \frac{2M_E}{r^3} + \frac{M_S}{R^3}$$

Using Newton's Method: $r_{n+1} = r_n - \frac{f(r_n)}{f'(r_n)}$.

n	r_n (m)	$f(r_n)$	$f'(r_n)$
0	1.2270000×10^8	-3.9643797×10^8	6.4673125
1	1.8399872×10^8	-5.4057907×10^8	1.8802288
2	2.7150533×10^8	-5.9619623×10^8	0.6824368
3	3.9186642×10^8	-5.1118318×10^8	0.3264024
4	5.4847849×10^8	-3.3101662×10^8	0.1773641
5	7.3510825×10^8	-1.6508358×10^8	0.0884333
6	9.3179294×10^8	-5.6729910×10^7	0.0327731
7	1.1048995×10^9	-1.1611867×10^7	0.0096825
8	1.2248233×10^9	–	–

The distance r corrected to 7 significant figures is:

$$r \approx 1.224823 \times 10^9 \text{ m}$$